

# Umer Cheema

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## Education

### University of Waterloo

Waterloo, ON

*BCFM, Honours Computer Science and Financial Management (Co-op)*

- Coursework: Financial Markets, Algorithm Design & Data Structures, Financial Reporting, Functional Programming, Calculus I & II, Linear Algebra, Discrete Math

## Experience

### Summer Analyst, Treasury & Forex Group

Lahore, PK

*MCB Bank Limited*

*Jul 2026 – Sep 2026*

- Built and shipped a **Bloomberg-style** AI news terminal for the treasury desk that scraped Pakistan's three largest financial news sources and scored relevance and sentiment, replacing a manual routine that cost the **30+ person desk** about **15 analyst-hours/day**.
- Analyzed the bank's daily **FX blotter** (NOP, Nostro, forward points, reval rates) across a book carrying **\$100M+** in live USD exposure.
- Contributed to daily **FX swap point** pricing calculations and built working knowledge of **options, futures, and forward** pricing mechanics on a desk managing nine-figure (USD) exposure.
- Tracked daily **CRR/SLR** ratios against regulatory thresholds and analyzed **T-bill and Pakistan Investment Bond (PIB)** auction data and yield movements for the money market team within Treasury.

### Software Engineer Intern

Lahore, PK

*MCB Bank Limited*

*May 2026 – Jul 2026*

- Built the retrieval and reasoning core of **MCB Live's** AI chatbot: a custom **RAG pipeline** (chunking, vector store, retrieval logic) using **MiniLM** embeddings and **Groq** for LLM reasoning. Projected to save **200M+ PKR ( 7 figures USD)** annually and cut support agent load by **35%**.
- Designed **guardrails** preventing PII exposure and built a **PM-facing dev panel** letting non-technical product managers tune prompts, temperature/warmth, and guardrails without code, plus **multi-language** support.
- Designed and ran a **61-case QA suite** (standard queries + PII/language handling) on the chatbot, uncovering critical backend security and architecture gaps that the bank adopted for remediation.

### Software Engineer Intern

Toronto, ON

*Krownie · AI-Powered Voice Interview Platform*

*May 2025 – Aug 2025*

- Engineered real-time voice interview pipelines using **OpenAI API** and **Retell AI** on serverless **AWS Lambda/S3** infrastructure with **Supabase PostgreSQL**, handling prompt chaining and session state for a production candidate screening tool.
- Cut interview failure rate by **30%** by hardening prompt logic and edge-case handling in speech transcription and LLM response parsing, improving end-to-end session completion.

### Finance Intern

Remote

*Katsah.com · E-Commerce Platform*

*May 2024 – Aug 2024*

- Wrote **SQL** queries to diagnose checkout funnel drop-off (pinpointing leakage costing **50–60%** of potential conversions) and built **Excel** variance models isolating pricing/marketing spend as revenue drivers; both directly informed leadership budget decisions.

## Projects

### Machine Learning Robo-Advisory System

*Python, Pandas, NumPy, yfinance, Scikit-learn*

- Built a monthly stock-selection engine across **50 large-cap US equities** using an **LR/RF/GBM ensemble** with 10 engineered features (momentum, RSI, volatility, beta); walk-forward validated across **23 out-of-sample periods**.
- Applied **fractional Kelly sizing** (25% fraction, 20% cap) and **GMM regime detection** to scale exposure at 100/75/50% (regime overlay drove **40% of alpha**), delivering **+67.7% total return vs. +46.3% SPY** (Feb 2023–Dec 2024), **Sharpe 1.89**, max drawdown **8%**.

## Technical Skills

**Languages:** Python, C, JavaScript, TypeScript, SQL, Java, C++, HTML/CSS

**Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, SciPy, Statsmodels, TensorFlow, LangChain

**Frameworks:** Next.js, React, FastAPI, REST API Node.js, Flask

**Systems:** AWS (Lambda, S3), Supabase, Git, Linux, Jupyter, VS Code, PowerBI, Excel, Azure

**Finance:** Financial Markets & FX/Fixed Income, Risk-Adjusted Portfolio Evaluation, Financial Reporting, GAAP, ASPE, IFRS